Abdullah Shaukat

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Education

Drexel University — Philadelphia, PA
B.S. in Computer Science | Minors: Data Science, Economics — *Expected Graduation: June 2029*

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, Java, C, R, HTML, Visual Basic
Frameworks & Libraries: FastAPI, Next.js (React), Streamlit, Pandas, NumPy, Scikit-learn, XGBoost, SHAP
ML/AI: PyTorch, TensorFlow, Reinforcement Learning (DQN), NLP, Sentence-BERT, TF-IDF, LLM APIs (Claude)
Data & Tools: PostgreSQL/pgvector, Redis, Docker, Git, SUMO, Jupyter, Power BI, Tableau, Excel, STATA

Experience

Researcher | *Python, SUMO/TraCI, Pandas, Reinforcement Learning (DQN, KNN-TD)* — Sept 2026 – Present
VIP: Detecting Cyberattacks in Traffic Systems, Drexel University — *Philadelphia, PA*

- Simulated a city intersection in **SUMO** and Python to study how **hacked sensors** disrupt smart traffic lights
- Tested **DQN** and **KNN-TD** reinforcement learning on traffic lights; KNN-TD cut delays **10%** vs. fixed timing
- Generated **30 hours** of clean traffic data across **6 scenarios**, from light traffic to rush hour
- Analyzed **100K+ seconds** of sensor data in **Pandas**, defining limits that expose faked sensor readings
- Building an **anomaly detector** to flag hacked traffic-sensor data across **3 simulated attack types**

Researcher | *Python, Pandas, NumPy, NetworkX, Matplotlib and OLS regression* — Apr – Sep 2026
Data Engineering at CAM lab, Drexel University — *Philadelphia, PA*

- Built a spatiotemporal simulator modeling flows for **1.1M+ daily journeys** across a 147-station metro system
- Optimized shortest-path calculations, reducing unique path pairs from **1.1M to <22K**
- Accelerated simulation runtimes by **over 90%** by eliminating redundant **Dijkstra** algorithm path executions
- Engineered trip time throughout the day via **linear regression**, decomposing it into wait, speed, and dwell

Operations Assistant — Sep 2025 – Mar 2026
Ronald McDonald House Charities — *Philadelphia, PA*

- Built Excel inventory systems across 4+ areas, lifting tracking accuracy by **30%** and cutting over-ordering by **20%**
- Analyzed 1000+ family stays over 5 years, building a **demand-peak dashboard** that reshaped capacity planning
- Designed a **donation analytics dashboard** on **$1M+** in gifts, surfacing patterns that retargeted fundraising

Projects

PaperMatch | *Python, FastAPI, PostgreSQL/pgvector, Next.js, Claude API, OpenAlex* — Sep 2026 – Present

- Designing an AI research tool that finds papers keyword search misses by following citation graphs
- Merging paper data from 4 scholarly APIs (OpenAlex, arXiv, Semantic Scholar, Crossref), removing duplicates
- Building a multi-hop citation traversal that reaches target papers only findable 3 hops from the query

CreditLens | *Python, Pandas, Scikit-learn, XGBoost, SHAP, Streamlit* — Aug 2026 – Present

- Built an ML credit-risk application using **30,000+** borrower records to predict probability of credit-card default
- Engineered financial and repayment features, including credit utilization, payment ratios, and delinquency history
- Compared **Logistic Regression** & **XGBoost** using ROC-AUC, precision, recall, and F1 to select the best model
- Integrated **SHAP** and **Streamlit** to visualize borrower risk factors and simulate changes to credit profiles

ScentSync | *Python, Pandas, Scikit-learn, Sentence-BERT, Streamlit, NLP* — Mar 2026 – Present

- Developed and deployed a **Streamlit** web app serving real-time fragrance recommendations and dupe results
- Built a dual-model search engine combining **TF-IDF** and **Sentence-BERT** for keyword & contextual matching
- Cleaned **24,000+ fragrances** in **Pandas** and engineered 6 note/accord columns into natural-language templates
- Designed a multi-stage dupe detector using **cosine similarity** to surface budget alternatives to luxury scents

NYS Food Service Compliance & Inspection Database | *SQL, RedGate Data Modeler* — Mar – May 2026

- Engineered a relational database ingesting **21,000+** NYS food safety inspection records into a 13-table schema
- Designed a **3NF** schema and **SQL ETL pipeline** in **RedGate Data Modeler**

Leadership & Activities

Leadership: Treasurer, Drexel PSA (Feb 2025 – Present) | FIFA, Host City Ambassador (Mar 2026 - Present)
Activities: Desk Assistant, Drexel University | Drexel Data Science Club | Drexel CyberDragons | Drexel AI Club